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Specialty: Cardiovascular / Pulmonary
Sample Name: Complex Cyanotic Congenital Heart Disease

HISTORY:, The patient is a 10-1/2-year-old born with asplenia syndrome with a complex cyanotic congenital heart disease characterized by dextrocardia bilateral superior vena cava, complete atrioventricular septal defect, a total anomalous pulmonary venous return to the right-sided atrium, and double-outlet to the right ventricle with malposed great vessels, the aorta being anterior with a severe pulmonary stenosis. He had undergone staged repair beginning on 04/21/1997 with a right modified Blalock-Taussig shunt followed on 09/02/1999 with a bilateral bidirectional Glenn shunt, and left pulmonary artery to main pulmonary artery pericardial patch augmentation. These procedures were performed at Medical College Hospital. Family states that they moved to the United States. Evaluation at the Children's Hospital earlier this year demonstrated complete occlusion of the right bidirectional Glenn shunt as well as occlusion of the proximal right pulmonary artery. He was also found to have elevated Glenn pressures at 22 mmHg, transpulmonary gradient axis of 14 mmHg. The QP:QS ratio of 0.6:1. A large decompressing venous collateral was also appreciated. The patient was brought back to cardiac catheterization in an attempt to reconstitute the right caval pulmonary anastomosis and to occlude the venous collateral vessel.,DESCRIPTION OF PROCEDURE: , After sedation and local Xylocaine anesthesia, the patient was prepped and draped. Cardiac catheterization was performed as outlined in the attached continuation sheets. Vascular entry was by percutaneous technique, and the patient was heparinized. Monitoring during the procedure included continuous surface ECG, continuous pulse oximetry, and cycled cuff blood pressures, in addition to intravascular pressures.,Using a 6-French sheath, a 6-French wedge catheter was inserted in the right femoral vein and advanced from the inferior vena cava into the right-sided atrium pulmonary veins and the right ventricle.,Using a 6-French sheath, a 5-French pigtail catheter was inserted into the right femoral artery and advanced retrograde to the descending aorta and ascending aorta. A separate port of arterial access was obtained in the left femoral artery utilizing a 5-French sheath.,Percutaneous access into the right jugular vein was attempted, but unsuccessful. Ultrasound on the right neck demonstrated a complete thrombosis of the right internal jugular vein. Using percutaneous technique and a 5-French sheath, 5-French wedge catheter was inserted into the left internal jugular vein and advanced along the left superior vena cava across the left caval-pulmonary anastomosis into the main pulmonary artery and left pulmonary artery with aid of guidewire. This catheter then also advanced into the bridging innominate vein. The catheter was then exchanged over wire for a 4-French Bernstein catheter, which was advanced to the blind end of the right superior vena cava. A balloon wedge angiogram of the right lower pulmonary vein demonstrated back filling of a small right lower pulmonary artery. There was no vascular continuity to the stump of the right Glenn. The jugular venous catheter and sheaths were exchanged over a wire

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for a 6-French flexor sheath, which was advanced to the proximal right superior vena cava. The Bernstein catheter was then reintroduced using a Terumo guidewire. Probing of the superior vena cava facilitated access into the right lower pulmonary artery. The angiogram in the right pulmonary artery showed a diminutive right lower pulmonary artery and severe long segment proximal stenosis. The distal pulmonary measured approximately 5.5 to 60 mm in diameter with a long segment stenosis measuring approximately 31 mm in length. The length of the obstruction was balloon dilated using ultra-thin SD 4 x 2 cm balloon catheter with complete disappearance of the waist. This facilitated advancement of a flexor sheath into the proximal portion of the stenosis. A PG 2960 BPX Genesis stent premounted on a 6 mm OptiProbe. A balloon catheter was advanced across the area of narrowing and inflated with a near-complete disappearance of proximal waist. Angiogram demonstrated a good stent apposition to the caval wall. Further angioplasty was then performed utilizing an ultra-thin SDS 8 x 3 cm balloon catheter inflated to 19 atmospheres pressure with complete disappearance of a distinct proximal waist. Angiogram demonstrated wide patency of reconstituted right caval pulmonary anastomosis though there was no flow seen to the right upper pulmonary artery. The balloon wedge angiograms were then obtained in the right upper pulmonary veins suggesting the presence of right upper pulmonary artery and not contiguous with the right lower pulmonary artery. Bernstein catheter was advanced into the main pulmonary artery where a wire probing of the stump of the proximal right pulmonary artery facilitated access to the right upper pulmonary artery. Angiogram demonstrated severe long segment stenosis of the proximal right pulmonary artery. Angioplasty of the right pulmonary was then performed using the OptiProbe 6-mm balloon catheter inflated to 16 atmospheres pressure with disappearance of a distinct waist. Repeat angiogram showed improvement in caliber of right upper pulmonary artery with filling defect of the proximal right pulmonary artery. The proximal right pulmonary artery was then dilated and stent implanted using a PG 2980 BPX Genesis stent premounted on 8-mm OptiProbe balloon catheter and implanted with complete disappearance of the waist. Distal right upper pulmonary artery was then dilated and stent implanted utilizing a PG 1870 BPX Genesis stent premounted on 7-mm OptiProbe balloon catheter. Repeat angiograms were then performed. Attention was then directed to the large venous collateral vessel arising from the left superior vena cava with a contrast filling of a left-sided azygos vein. A selective angiogram demonstrated a large azygos vein of the midsection measuring approximately 9.4 mm in diameter. An Amplatzer 12 mm vascular plug was loaded on the delivery catheter and advanced through the flexor sheath into the azygos vein. Once stable device was confirmed, the device was released from the delivery catheter. The 4-French Bernstein catheter was then reintroduced and 5 inch empirical 0.038 inch, 10 cm x 8 mm detachable coils were then

implanted above the vascular plug filling the proximal azygos vein. A pigtail catheter was then introduced into the left superior vena cava for final angiogram.,Flows were calculated by the Fick technique using an assumed oxygen consumption and contents derived from Radiometer Hemoximeter saturations and hemoglobin capacity.,Cineangiograms were obtained with injection of the coronary sinus of pulmonary veins, the innominate vein, superior vena cava, the main pulmonary artery, and azygos vein.,After angiography, two normal-appearing renal collecting systems were visualized. The catheters and sheaths were removed and topical pressure applied for hemostasis. The patient was returned to the recovery room in satisfactory condition. There were no complications.,DISCUSSION:, Oxygen consumption was assumed to be normal mixed venous saturation, but was low due to systemic arterial desaturation of 79%. The pulmonary veins were fully saturated with partial pressure of oxygen ranging between 120 and 169 mmHg in 30% oxygen. Remaining saturations reflected complete admixture. There was increased saturation in the left pulmonary artery due to aortopulmonary collateral flow. Phasic right atrial pressures were normal with an A-wave somewhat to the normal right ventricular end-diastolic pressure of 9 mmHg. Left ventricular systolic pressure was systemic with no outflow obstruction to the ascending aorta. Phasic ascending, descending pressures were similar and normal. Mean Glenn pressures at initiation of the case were slightly elevated at 14 mmHg with a transpulmonary gradient of 9 mmHg. The calculated systemic flow was a normal pulmonary flows reduced with a QP:QS ratio of 0.6:1. The pulmonary vascular resistance was elevated at 4.4 Woods units. Following stent implantation in the right caval pulmonary anastomosis and right pulmonary artery, there was a slight increase in the Glenn venous pressures to 16 mmHg. Following embolization of the azygos vein, there was increase in systemic arterial saturation to 84% and increase in mixed venous saturation. There was similar increase in Glenn pressures to 28 mmHg with a transpulmonary gradient of 14 mmHg. There was an increase in arterial pressure. The calculated systemic flow increased from 3.1 liters /minute/meter squared to 4.3 liters/minute/meter squared. Angiogram within the innominate vein following stent implantation demonstrated appropriate stent position without significant distortion of the innominate vein or proximal cava. There appeared unobstructed contrast flow to the right lower pulmonary artery of a 1-mmHg mean pressure gradient. There was absence of contrast filling of the right middle and right upper pulmonary artery. Final angiogram with a contrast injection in the left superior vena cava showed a forward flow through the right Glenn, a good contrast filling of the right lower pulmonary artery, and a widely patent left Glenn negative contrast washout of the proximal right pulmonary artery and left pulmonary artery presumably due to aortopulmonary collateral flow. Contrast injection within the right upper pulmonary artery following the stent implantation

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demonstrated widely patent proximal right pulmonary artery along the length of the implanted stents though with retrograde contrast flow.,INITIAL DIAGNOSES: ,1. Asplenia syndrome.,2. Dextrocardia bilateral superior vena cava.,3. Atrioventricular septal defect.,4. Total anomalous pulmonary venous return to the right-sided atrium.,5. Double outlet right ventricle with malposed great vessels.,6. Severe pulmonary stenosis.,7. Separate hepatic venous drainage into the atria.,PRIOR SURGERIES AND INTERVENTIONS: ,1. Right modified Blalock-Taussig shunt.,2. Bilateral bidirectional Glenn shunt.,3. Patch augmentation of the main pulmonary to left pulmonary artery.,CURRENT DIAGNOSES: ,1. Obstructed right caval pulmonary anastomosis.,2. Obstructed right proximal pulmonary artery.,3. Venovenous collateral vessel.,CURRENT INTERVENTION: ,1. Balloon dilation of the right superior vena cava and stent implantation.,2. Balloon dilation of the proximal right pulmonary artery, stent implantation.,3. Embolization of venovenous collateral vessel.,MANAGEMENT: , The case will be discussed in Combined Cardiology Cardiothoracic Surgery case conference. A repeat catheterization is recommended in 3 months to assess for right pulmonary artery growth and to assess candidacy for Fontan completion. The patient will be maintained on anticoagulant medications of aspirin and Plavix. Further cardiology care will be directed by Dr. X.

Specialty: Chiropractic
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DATE OF INJURY : October 4, 2000,DATE OF EXAMINATION : September 5, 2003,EXAMINING PHYSICIAN : X Y, MD,Prior to the beginning of the examination, it is explained to the examinee that this examination is intended for evaluative purposes only, and that it is not intended to constitute a general medical examination. It is explained to the examinee that the traditional doctor-patient relationship does not apply to this examination, and that a written report will be provided to the agency requesting this examination. It has also been emphasized to the examinee that he should not attempt any physical activity beyond his tolerance, in order to avoid injury.,CHIEF COMPLAINT: ,Aching and mid back pain.,HISTORY OF PRESENT INJURY: , Based upon the examinee's perspective: ,Mr. Abc is a 52-year-old self-employed, independent consultant for DEMILEE-USA. He is also a mechanical engineer. He reports that he was injured in a motor vehicle accident that occurred in October 4, 2000. At that time, he was employed as a purchasing agent for IBIKEN-USA. On the date of the motor vehicle accident, he was sitting in the right front passenger's seat, wearing seat and shoulder belt safety harnesses, in his 1996 or 1997 Volvo 850 Wagon automobile driven by his son. The vehicle was completely stopped and was "slammed from behind" by a van. The police officer, who responded to the accident, told Mr. Abc that the van was probably traveling at approximately 30 miles per hour at the time of impact.,During the impact, Mr. Abc was restrained in the seat and did not contact the interior surface of the vehicle. He experienced immediate mid back pain. He states that the Volvo automobile sustained approximately $4600 in damage.,He was transported by an ambulance, secured by a cervical collar and backboard to the emergency department. An x-ray of the whole spine was obtained, and he was evaluated by a physician's assistant. He was told that it would be "okay to walk." He was prescribed pain pills and told to return for reevaluation if he experienced increasing pain.,He returned to the Kaiser facility a few days later, and physical therapy was prescribed. Mr. Abc states that he was told that "these things can take a long time." He indicates that after one year he was no better. He then states that after two years he was no better and worried if the condition would never get better.,He indicates he saw an independent physician, a general practitioner, and an MRI was ordered. The MRI study was completed at ABCD Hospital. Subsequently, Mr. Abc returned and was evaluated by a physiatrist. The physiatrist reexamined the original thoracic spine x-rays that were taken on October 4, 2000, and stated that he did not know why the radiologist did not originally observe vertebral compression fractures. Mr. Abc believes that he was told by the physiatrist that it involved either T6-T7 or T7-T8.,Mr. Abc reports that the physiatrist told him that little could be done besides participation in core strengthening. Mr. Abc describes his current exercise regimen, consisting of cycling, and it was deemed to be adequate. He was told, however, by the physiatrist that he could also try a Pilates type of

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core exercise program.,The physiatrist ordered a bone scan, and Mr. Abc is unsure of the results. He does not have a formal follow up scheduled with Kaiser, and is awaiting re-contact by the physiatrist.,He denies any previous history of symptomatology or injuries involving his back.,CURRENT SYMPTOMS: ,He reports that he has the same mid back pain that has been present since the original injury. It is located in the same area, the mid thoracic spine area. It is described as a pain and an ache and ranges from 3/10 to 6/10 in intensity, and the intensity varies, seeming to go in cycles. The pain has been staying constant.,When I asked whether or not the pain have improved, he stated that he was unable to determine whether or not he had experienced improvement. He indicates that there may be less pain, or conversely, that he may have developed more of a tolerance for the pain. He further states that "I can power through it." "I have learned how to manage the pain, using exercise, stretching, and diversion techniques." It is primary limitation with regards to the back pain involves prolonged sitting. After approximately two hours of sitting, he has required to get up and move around, which results in diminishment of the pain. He indicates that prior to the motor vehicle accident, he could sit for significantly longer periods of time, 10 to 12 hours on a regular basis, and up to 20 hours, continuously, on an occasional basis.,He has never experienced radiation of the pain from the mid thoracic spine, and he has never experienced radicular symptoms of radiation of pain into the extremities, numbness, tingling, or weakness.,Again, aggravating activities include prolonged sitting, greater than approximately two hours.,Alleviating activities include moving around, stretching, and exercising. Also, if he takes ibuprofen, it does seem to help with the back pain.,He is not currently taking medications regularly, but list that he takes occasional ibuprofen when the pain is too persistent.,He indicates that he received several physical therapy sessions for treatment, and was instructed in stretching and exercises. He has subsequently performed the prescribed stretching and exercises daily, for nearly three years.,With regards to recreational activities, he states that he has not limited his activities due to his back pain.,He denies bowel or bladder dysfunction.,FILES REVIEW: ,October 4, 2000: An ambulance was dispatched to the scene of a motor vehicle accident on South and Partlow Road. The EMS crew arrived to find a 49-year-old male sitting in the front passenger seat of a vehicle that was damaged in a rear-end collision and appeared to have minimal damage. He was wearing a seatbelt and he denied loss of consciousness. He also denied a pertinent past medical history. They noted pain in the lower cervical area, mid thoracic and lumbar area. They placed him on a backboard and transported him to Medical Center.,October 4, 2000: He was seen in the emergency department of Medical Center. The provider is described as "unknown." The history from the patient was that he was the passenger in the front seat of a car that was stopped and rear-ended. He stated that he did

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not exit the car because of pain in his upper back. He reported he had been wearing the seatbelt and harness at that time. He denied a history of back or neck injuries. He was examined on a board and had a cervical collar in place. He was complaining of mid back pain. He denied extremity weakness. Sensory examination was intact. There was no tenderness with palpation or flexion in the neck. The back was a little tender in the upper thoracic spine area without visible deformity. There were no marks on the back. His x-ray was described as "no acute bony process." Listed visit diagnosis was a sprain-thoracic, and he was prescribed hydrocodone/acetaminophen tablets and Motrin 800 mg tablets.,October 4, 2000: During the visit, a Clinician's Report of Disability document was signed by Dr. M, authorizing time loss from October 4, 2000, through October 8, 2000. The document also advised no heavy lifting, pushing, pulling, or overhead work for two weeks. During this visit, a thoracic spine x-ray series, two views, was obtained and read by Dr. JR. The findings demonstrate no evidence of acute injury. No notable arthritic findings. The pedicles and paravertebral soft tissues appear unremarkable.,November 21, 2000: An outpatient progress note was completed at Kaiser, and the clinician of record was Dr. H. The history obtained documents that Mr. Abc continued to experience the same pain that he first noted after the accident, described as a discomfort in the mid thoracic spine area. It was non-radiating and described as a tightness. He also reported that he was hearing clicking noises that he had not previously heard. He denied loss of strength in the arms. The physical examination revealed good strength and normal deep tendon reflexes in the arms. There was minimal tenderness over T4 through T8, in an approximate area. The visit diagnosis listed was back pain. Also described in the assessment was residual pain from MVA, suspected bruised muscles. He was prescribed Motrin 800 mg tablets and an order was sent to physical therapy. Dr. N also documents that if the prescribed treatment measures were not effective, then he would suggest a referral to a physiatrist. Also, the doctor wanted him to discuss with physical therapy whether or not they thought that a chiropractor would be beneficial.,December 4, 2000: He was seen at Kaiser for a physical therapy visit by Philippe Justel, physical therapist. The history obtained from Mr. Abc is that he was not improving. Symptoms described were located in the mid back, centrally. The examination revealed mild tenderness, centrally at T3-T8, with very poor segmental mobility. The posture was described as rigid T/S in flexion. Range of motion was described as within normal limits, without pain at the cervical spine and thoracic spine. The plan listed included two visits per week for two weeks, for mobilization. It is also noted that the physical therapist would contact the MD regarding a referral to a chiropractor.,December 8, 2000: He was seen at Kaiser for a physical therapy visit by Mr. Justel. It was noted that the subjective category of the document revealed that there was no real change. It was noted that Mr. Abc tolerated the treatment well

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and that he was to see a chiropractor on Monday.,December 11, 2000: He presented to the Chiropractic Wellness Center. There is a form titled 'Chiropractic Case History,' and it documents that Mr. Abc was involved in a motor vehicle accident, in which he was rear-ended in October. He has had mid back pain since that time. The pain is worsened with sitting, especially at a computer. The pain decreases when he changes positions, and sometimes when he walks. Mr. Abc reports that he occasionally takes 800 mg doses of ibuprofen. He reported he went to physical therapy treatment on two occasions, which helped for a few hours only. He did report that he had a previous history of transitory low back pain.,During the visit, he completed a modified Oswestry Disability Questionnaire, and a WC/PI Subjective Complaint Form. He listed complaints of mid and low back pain of a sore and aching character. He rated the pain at grade 3-5/10, in intensity. He reported difficulty with sitting at a table, bending forward, or stooping. He reported that the pain was moderate and comes and goes.,During the visit at the Chiropractic Wellness Center, a spinal examination form was completed. It documents palpation tenderness in the cervical, thoracic, and lumbar spine area and also palpation tenderness present in the suboccipital area, scalenes, and trapezia. Active cervical range of motion measured with goniometry reveals pain and restriction in all planes. Active thoracic range of motion measured with inclinometry revealed pain and restriction in rotation bilaterally. Active lumbosacral range of motion measured with inclinometry reveals pain with lumbar extension, right lateral flexion, and left lateral flexion.,December 11, 2000: He received chiropractic manipulation treatment, and he was advised to return for further treatment at a frequency of twice a week.,December 13, 2000: He returned to the Chiropractic Wellness Center to see Joe Smith, DC, and it is documented that his middle back was better.,December 13, 2000: A personal injury patient history form is completed at the Chiropractic Wellness Center. Mr. Abc reported that on October 4, 2000, he was driving his 1996 Volvo 850 vehicle, wearing seat and shoulder belt safety harnesses, and completely stopped. He was rear-ended by a vehicle traveling at approximately 30 miles per hour. The impact threw him back into his seat, and he felt back pain and determined that it was not wise to move about. He reported approximate damage to his vehicle of $4800. He reported continuing mid and low back pain, of a dull and semi-intense nature. He reported that he was an export company manager for IBIKEN-USA, and that he missed two full days of work, and missed 10-plus partial days of work. He stated that he was treated initially after the motor vehicle accident at Kaiser and received painkillers and ibuprofen, which relieved the pain temporarily. He specifically denied ever experiencing similar symptoms.,December 26, 2000: A no-show was documented at the Chiropractic Wellness Center.,April 5, 2001: He received treatment at the Chiropractic Wellness Center. He reported that two weeks previously, his mid back pain had

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worsened.,April 12, 2001: He received chiropractic treatment at the Chiropractic Wellness Center.,April 16, 2001: He did not show up for his chiropractic treatment.,April 19, 2001: He did not show up for his chiropractic treatment.,April 26, 2001: He received chiropractic manipulation treatment at the Chiropractic Wellness Center. He reported that his mid back pain increased with sitting at the computer. At the conclusion of this visit, he was advised to return to the clinic as needed.,September 6, 2002: An MRI of the thoracic spine was completed at ABCD Hospital and read by Dr. RL, radiologist. Dr. D noted the presence of minor anterior compression of some mid thoracic vertebrae of indeterminate age, resulting in some increased kyphosis. Some of the mid thoracic discs demonstrate findings consistent with degenerative disc disease, without a significant posterior disc bulging or disc herniation. There are some vertebral end-plate abnormalities, consistent with small Schmorl's nodes, one on the superior aspect of T7, which is compressed anteriorly, and on the inferior aspect of T6.,May 12, 2003: He was seen at the Outpatient Clinic by Dr. L, internal medicine specialist. He was there for a health screening examination, and listed that his only complaints are for psoriasis and chronic mid back pain, which have been present since a 2000 motor vehicle accident. Mr. Abc reported that an outside MRI showed compression fractures in the thoracic spine. The history further documents that Mr. Abc is an avid skier and volunteers on the ski patrol. The physical examination revealed that he was a middle-aged Caucasian male in no acute distress. The diagnosis listed from this visit is back pain and psoriasis. Dr. L documented that he spent one hour in the examination room with the patient discussing what was realistic and reasonable with regard to screening testing. Dr. L also stated that since Mr. Abc was experiencing chronic back pain, he advised him to see a physiatrist for evaluation. He was instructed to bring the MRI to the visit with that practitioner.,June 10, 2003: He was seen at the Physiatry Clinic by Dr. R, physiatrist. The complaint listed is mid back pain. In the subjective portion of the chart note, Dr. R notes that Mr. Abc is involved in the import/export business, and that he is physically active in cycling, skiing, and gardening. He is referred by Dr. L because of persistent lower thoracic pain, following a motor vehicle accident, on October 4, 2000. Mr. Abc told Dr. R that he was the restrained passenger of a vehicle that was rear-ended at a moderate speed. He stated that he experienced immediate discomfort in his thoracic spine area without radiation. He further stated that thoracic spine x-rays were obtained at the Sunnyside Emergency Room and read as normal. It is noted that Mr. Abc was treated conservatively and then referred to physical therapy where he had a number of visits in late of 2002 and early 2003. No further chart entries were documented about the back problem until Mr. Abc complained to Dr. L that he still had ongoing thoracic spine pain during a visit the previous month. He obtained an MRI, out of pocket, at ABCD Hospital and stated that he paid

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$1100 for it. Dr. R asked to see the MRI and was told by Mr. Abc that he would have to reimburse or pay him $1100 first. He then told the doctor that the interpretation was that he had a T7 and T8 compression fracture. Mr. Abc reported his improvement at about 20%, compared to how he felt immediately after the accident. He described that his only symptoms are an aching pain that occurs after sitting for four to five hours. If he takes a break from sitting and walks around, his symptoms resolve. He is noted to be able to bike, ski, and be active in his garden without any symptoms at all. He denied upper extremity radicular symptoms. He denied lower extremity weakness or discoordination. He also denied bowel or bladder control or sensation issues. Dr. R noted that Mr. Abc was hostile about the Kaiser health plan and was quite uncommunicative, only reluctantly revealing his history. The physical examination revealed that he moved about the examination room without difficulty and exhibited normal lumbosacral range of motion. There was normal thoracic spine motion with good chest expansion. Neurovascular examination of the upper extremities was recorded as normal. There was no spasticity in the lower extremities. There was no tenderness to palpation or percussion up and down the thoracic spine. Dr. R reviewed the thoracic spine films and noted the presence of "a little compression of what appears to be T7 and T8 on the lateral view." Dr. R observed that this was not noted on the original x-ray interpretation. He further stated that the MRI, as noted above, was not available for review. Dr. R assessed that Mr. Abc was experiencing minimal thoracic spine complaints that probably related to the motor vehicle accident three years previously. The doctor further stated that "the patient's symptoms are so mild as to almost not warrant intervention." He discussed the need to make sure that Mr. Abc's workstation was ergonomic and that Mr. Abc could pursue core strengthening. He further recommended that Mr. Abc look into participation in a Pilates class. Mr. Abc was insistent, so Dr. R made plans to order a bone scan to further discriminate the etiology of his symptoms. He advised Mr. Abc that the bone scan results would probably not change treatment. As a result of this visit, Dr. R diagnosed thoracic spine pain (724.1) and ordered a bone scan study.

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Sample Name: Neurologic Consultation - 5

HISTORY OF PRESENT ILLNESS: ,The patient is a 79-year-old right-handed man who reports that approximately one and a half years ago, he fell down while walking in the living room from the bedroom. At that time, he reports both legs gave away on him and he fell. He reported that he had some lightheadedness just before he fell and was slightly confused, but was aware of what was happening around him. He was able to get up shortly after falling and according to the patient and his son, subsequently returned back to normal.,He was then well until the 3rd of July 2008 when his legs again gave way on him. This was not preceded by lightheadedness. He was rushed to the hospital and was found to have pneumonia, and the fall was blamed on the pneumonia. He started using a walker from that time, prior to that he was able to walk approximately two miles per day. He again had a fall in August of 2008 after his legs gave way. Again, there was no lightheadedness associated with this. He was again found to have pneumonia and again was admitted to hospital after which he went to rehabilitation and was able to use his walker again after this. He did not, however, return to the pre-July baseline. In October of 2008, after another fall, he was found to have pneumonia again and shingles. He is currently in a Chronic Rehabilitation Unit. He cannot use a walker and uses a wheelchair for everything. He states that his hands have been numb, involving all the fingers of both hands for the past three weeks. He is also losing muscle bulk in his hands and has noticed some general weakness of his hands. He does, however, note that strength in his hands has not been normal since July 2008, but it is clearly getting worse. He has been aware of some fasciculations in his legs starting in August 2008, these are present both in the lower legs and the thighs. He does not report any cramps, problems with swallowing or problems with breathing. He reports that he has had constipation alternating with diarrhea, although there has been no loss of control of either his bowel or bladder. He has had some problems with blood pressure drops, and does feel presyncopal when he stands. He also reports that he has no feeling in his feet, and that his feet feel like sponges. This has been present for about nine months. He has also lost joint position sense in his feet for approximately nine months.,PAST MEDICAL HISTORY:,1. Pneumonia. He has had recurrent episodes of pneumonia, which started at approximately age 20. These have been treated repeatedly over the years, and on average he has tended to have an episode of pneumonia once every five years, although this has been far more frequent in the past year. He is usually treated with antibiotics and then discharged. There is no known history of bronchiectasis, inherited lung disease or another chronic pulmonary cause for the repeated pneumonia.,2. He has had a catheter placed for urinary retention, his urologist has told him that he thinks that this may be due to prostate enlargement. The patient does not have any history of diabetes and does not report any other medical problems. He has lost approximately 18

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Sample Name: Neurologic Consultation - 5

pounds in the past month.,3. He had an appendectomy in the 1940s.,4. He had an ankle resection in 1975.,SOCIAL HISTORY: ,The patient stopped smoking 27 years ago, he smoked approximately two packs a day with combined cigarettes and cigars. He has not smoked for the past 27 years. He hardly ever uses alcohol. He is currently retired.,FAMILY HISTORY: , There is no family history of neuropathy, pes cavus, foot deformities, or neuromuscular diseases. His aunt has a history of type II diabetes.,CURRENT MEDICATIONS: , Fludrocortisone 0.1 mg p.o. q.d., midodrine 5 mg p.o. q.i.d., Cymbalta 30 mg p.o. per day, Prilosec 20 mg p.o. per day, Lortab 10 mg p.o. per day, Amoxil 500 mg p.o. per day, vitamin B12 1000 mcg weekly, vitamin D 1000 units per day, Metamucil p.r.n., enteric-coated aspirin once a day, Colace 200 mg p.o. q.d., Senokot three tablets p.o. p.r.n., Reglan 10 mg p.o. q.6h., Xanax 0.25 mg p.o. q.8h. p.r.n., Ambien 5 mg p.o. q.h.s. p.r.n. and Dilaudid 2 mg tablets p.o. q.3h. p.r.n., Protonix 40 mg per day, and Megace 400 mg per day.,ALLERGIES:, He has no medication or food allergies.,REVIEW OF SYSTEMS:, Please see the health questionnaire and clinical notes from today.,GENERAL PHYSICAL EXAMINATION:,VITAL SIGNS: BP was 137/60, P was 89, and his weight could not be measured because he was in a wheelchair. His pain score was 0.,APPEARANCE: No acute distress. He is pleasant and well-groomed.,HEENT: Atraumatic, normocephalic. No carotid bruits appreciated.,LUNGS: There were few coarse crackles in both lung bases.,CARDIOVASCULAR: Revealed a normal first and second heart sound, with no third or fourth heart sound and no murmurs. The pulse was regular and of normal volume.,ABDOMEN: Soft with no masses and normal bowel sounds. There were no carotid bruits.,EXTREMITIES: No contractures appreciated.,NEUROLOGICAL EXAM:,MSE: His orientation, language, calculations, 100-7 tests were all normal. There was atrophy and fasciculations in both the arms and legs.,CRANIAL NERVES: Cranial nerve examination was normal with the exception that there was some mild atrophy of his tongue and possible fasciculations. His palatal movement was normal and gag reflex was normal.,MOTOR: Strength was decreased in all muscle groups as follows: Deltoid 4/4, biceps 4+/4+, triceps 5/5, wrist extensors 4+/4+, finger extensors 4-/4-, finger flexors 4-/4-, interossei 4-/4-, hip flexors 4+/4+, hip extensors 4+/4+, knee extensors 4/4, and knee flexors 4/4. Foot dorsiflexion, plantar flexion, eversion, toe extension and toe flexion was all 0 to 1. There was atrophy in both hands and general atrophy of the lower limb muscles. The feet were both cold and showed dystrophic features. Fasciculations were present mainly in the hands. There was evidence of dysmetria and past pointing in the left hand.,REFLEXES: Reflexes were 0 in all sites in the arms and legs. The jaw reflex was 2+. Vibration was severely decreased at the elbow and wrist and was absent in the fingers. Vibration was absent in the toes and ankle bilaterally and was severely decreased at the knee. Joint position sense was absent in the toes and severely decreased in the fingers. Pin perception was absent in the feet and

was decreased to the upper thighs. Pin was decreased or absent in the fingers and decreased above the elbows. The same distribution of sensory loss was found with monofilament testing.,COORDINATION: Coordination was barely normal in the right hand. Rapid alternating movements were decreased in the left hand greater than the right hand. The patient was unable to stand and therefore gait, Romberg's test and balance could not be assessed.,DIAGNOSTIC STUDIES: , Previous diagnostic studies and patient reports. There were extensive patient reports, all of which were reviewed. A previous x-ray study of the lateral chest performed in October 2008 showed poor inspiration with basilar atelectasis and an infiltrate. An x-ray of the cervical, thoracic and lumbar spine showed some evidence of lumbar spinal stenosis. A CTA of the neck with and without contrast performed in November 2008 showed minor stenosis in the left carotid, a mild hard and soft plaque in the right carotid with approximately 55% stenosis. The posterior circulation showed a slightly dominant right vertebral artery with no stenosis. There was no significant stenosis, but there was minor extracranial stenosis noted. An MRI of the brain with and without contrast performed in November 2008 showed no evidence of an acute infarct, major vascular occlusion, and no abnormal enhancement with gadolinium administration. There was also no significant sinusitis or mastoiditis. This was an essentially normal brain MRI. A CBC performed in January 2009 showed an elevated white cell count of 11.3, a low red cell count of 3.43, elevated MCH of 32.4 and the rest of the study was normal. An electrolyte study performed in January 2009 showed a sodium which was low at 127, a calcium which was low at 8.3, and a low protein of 5.2 and albumin of 3.1. The glucose was 86. TSH performed in January 2009 was 1.57, which is within the normal range. Vitamin B12 was greater than a 1000, which is normal and the folate was 18.2, which was normal. A myocardial stress study performed in December 2008 showed normal myocardial perfusion with Persantine Cardiolite SPECT. The ECG was non-diagnostic. There was normal regional wall motion of the left ventricle. The left ventricular ejection fraction was 68%, which is within the normal range for males. A CT of the lumbar spine without contrast performed in December 2008 showed a broad-based disc bulge at L1-L2, L2-L3, L3-L4 and L4-L5. At L5-S1, in addition to the broad-based disc bulge, there was also an osteophyte complex and evidence of flavum hypertrophy without canal stenosis. There was severe bilateral neural foraminal stenosis at L5-S1 and moderate neural foraminal stenosis at L1-L4. An echocardiogram was performed in November 2008 and showed mild left atrial enlargement, normal left ventricular systolic function, mild concentric left ventricular hypertrophy, scleral degenerative changes in the aortic and mitral apparatus, mild mitral regurgitation, mild tricuspid regurgitation and mild to moderate aortic regurgitation.,DIAGNOSTIC IMPRESSION: ,The patient presents with a severe neuropathy with marked large fiber sensory as well as

motor findings. He is diffusely weak as well as atrophic in all muscle groups both in his upper and lower extremities, although he is disproportionately weak in his lower extremities. His proprioceptive and vibratory loss is severe in both the distal upper and lower extremities, signifying that he either has a severe sensory neuropathy or has involvement of the dorsal root ganglia. According to the history, which was carefully checked, the initial onset of these symptoms goes back one and a half years, although there has only been significant progression in his condition since July 2008. As indicated below, further diagnostic studies including a detailed nerve conduction and EMG test today showed evidence of a severe sensory, motor, and axonal neuropathy and in addition there was evidence of a diffuse polyradiculopathy. There was no involvement of the tongue on EMG. The laboratory testing as indicated below failed to show a specific cause for the neuropathy. We are still, however, waiting for the paraneoplastic antibodies, which were send out lab to the Mayo Clinic. This type of very severe sensorimotor neuropathy with significant proprioceptive loss may be seen in several conditions including peripheral nerve vasculitis due to a variety of disorders such as SLE, Sjogren's, rheumatoid arthritis, and mixed connective tissue disease. In addition, it may also be seen with certain toxins, particularly chemotherapeutic agents. The patient did not receive any of these. It may also be seen as part of a paraneoplastic syndrome. Although the patient does not have any specific clinical symptoms of a cancer, it is noted that he has had an 18-pound weight loss in the past month and does have a remote history of smoking. We have requested that he obtain a CT of his chest, abdomen and pelvis while he is in Acute Rehabilitation. The verbal reports of these possibly did not show any evidence of a cancer. We did also request that he obtain a gallium scan to see if there was any evidence of an unsuspected neoplasm. The patient did undergo a nerve and muscle biopsy, this was a radial nerve and biceps muscle biopsy from the left arm. This showed evidence of severe axonal loss. There was no evidence of a vasculitis. The vessels did show some mild intimal changes that would be consistent with atherosclerosis. There were a few perivascular changes; however, there was no clear evidence of a necrotizing vasculitis even on multiple sections. The muscle biopsy showed severe muscle fiber atrophy, with evidence of fiber grouping. Again, there was no evidence of inflammation or vasculitis. Evaluation so far has also shown no evidence of an amyloid neuropathy, no evidence of a monoclonal gammopathy, of sarcoidosis, and again there is no past history of a significant toxin or infective cause for the neuropathy. Specifically, there is no history of HIV exposure. We would await the results of the gallium scan and of the paraneoplastic antibodies to see if these are helpful in making a diagnosis. At this point, because of the severity and the axonal nature of the neuropathy, there is no specific therapy that will reverse the course of the

illness, unless we find a specific etiology that can be stopped or reversed. I have discussed these issues at length with the patient and with his son. We also addressed whether or not there might be a previously undiagnosed inherited neuropathy. I think this is unlikely given the short history and the rapid progression of the disorder.,There is also no family history that we can detect a neuropathy, and the patient does not have the typical phenotype for a chronic inherited neuropathy such as Charcot-Marie-Tooth disease type 2. However, since I have only seen the patient on one occasion and do not know what his previous examination showed two years ago, I cannot be certain that there may not have been the presence of a neuropathy preceding this.,PLAN:,1. Nerve conduction and EMG will be performed today. The results were indicated above.,2. The following laboratory studies were requested including electrolytes, CBC, thyroid function tests, B12, ANA, C-reactive protein, complement, cryoglobulins, double-stranded DNA antibodies, folate level, hemoglobin A1c, immunofixation electrophoresis, P-ANCA, C-ANCA, protein electrophoresis, rheumatoid factor, paraneoplastic antibody studies requested from the Mayo Clinic, B12. These studies showed minor changes, which included a low sodium level of 129 as previously noted, a low creatinine of 0.74, low calcium of 8.6, low total protein of 5.7. The B12 was greater than 2000. The immunoelectrophoresis, ANA, double-stranded DNA, ANCA, hemoglobin A1c, folate, cryoglobulins, complement, C-reactive protein were all normal or negative. The B12 level was greater than 2000. Liver function tests were normal. The glucose was 90. ESR was 10. Hemoglobin A1c was 5.5.,3. A left radial sensory and left biceps biopsy were requested and have been performed and interpreted as indicated above.,4. CT of chest, abdomen and pelvis.,5. Whole body gallium scan for evidence of an underlying neoplasm.,6. The patient will go to the Rehabilitation Facility for Acute Rehabilitation and Training.,7. We have not made any changes to his medication. He does have some mild orthostatic changes; however, he is adequately controlled with midodrine at a dose of 2.5 mg three times a day as needed up to 5 mg four times a day. Usually, he uses a lower dose of 2.5 three times a day to 5 mg three times a day.,8. Followup will be as determined by the family.

LABORATORY REPORT

Packet: 002

Analyte	Value	Units	Flag
Glucose	142	mg/dL	High
Hemoglobin	11.8	g/dL	Low
WBC	12.4	10*3/uL	High
Sodium	136	mmol/L	
Potassium	4.2	mmol/L	
Creatinine	1.4	mg/dL	High
BUN	24	mg/dL	High
Platelets	228	10*3/uL	
Hemoglobin A1c	7.6	%	High
TSH	2.1	mIU/L	